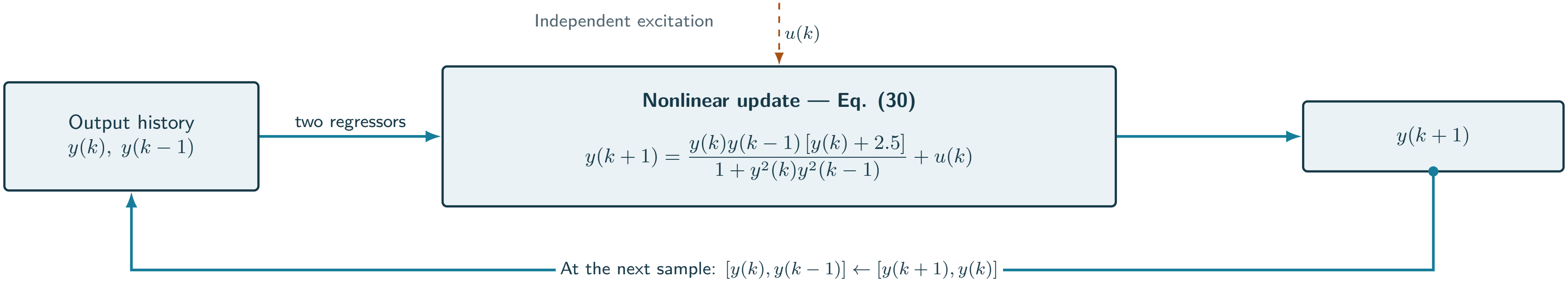
Artificial dynamic systems: two exact recurrences

Paper section 3.1: *Artificial system identification problems*

A. SISO: Narendra–Parthasarathy example 2

One input, one output; second-order recurrence



Independent excitation

$u(k)$

Output history
$y(k),\ y(k-1)$

two regressors

Nonlinear update — Eq. (30)

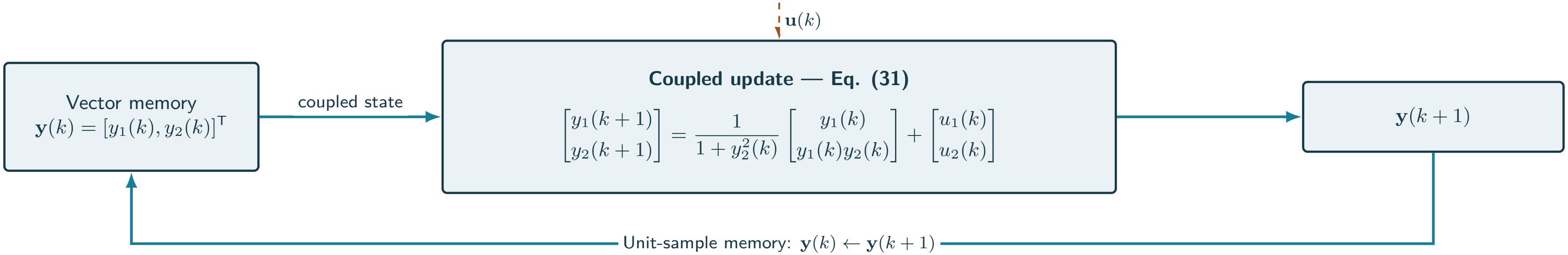
$$y(k+1) = \frac{y(k)y(k-1)\,[y(k)+2.5]}{1+y^2(k)y^2(k-1)} + u(k)$$

$y(k+1)$

At the next sample: $[y(k), y(k-1)] \leftarrow [y(k+1), y(k)]$

B. MIMO: Narendra–Parthasarathy example 4

Two inputs, two coupled outputs



$\mathbf{u}(k)$

Vector memory
$\mathbf{y}(k) = [y_1(k), y_2(k)]^\mathsf{T}$

coupled state

Coupled update — Eq. (31)

$$\begin{bmatrix} y_1(k+1) \\ y_2(k+1) \end{bmatrix} = \frac{1}{1+y_2^2(k)} \begin{bmatrix} y_1(k) \\ y_1(k)y_2(k) \end{bmatrix} + \begin{bmatrix} u_1(k) \\ u_2(k) \end{bmatrix}$$

$\mathbf{y}(k+1)$

Unit-sample memory: $\mathbf{y}(k) \leftarrow \mathbf{y}(k+1)$

Identification excitation

Article: 10^5 i.i.d. samples.
SISO: $u \sim \mathcal{U}[-2, 2]$.
MIMO: each $u_i \sim \mathcal{U}[-1, 1]$.

Validation excitation

SISO: $u(k) = \sin(2\pi k/25)$.
MIMO: $u_1(k) = \sin(2\pi k/25)$,
$u_2(k) = \cos(2\pi k/25)$.

Evaluation contract

Distinguish measured-history one-step prediction from recursive free-run prediction. Declare initialization, sample count and seed.

Blue arrows: numerical state/history propagation. Dashed orange arrows: independent excitation. These artificial identification problems have no implied physical plant. Repository zero-history initialization and random seeds are declared experimental choices.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Data resources: from records to a reproducible benchmark

Paper section 3.2: *Data repositories for system identification and regression*

Seven resource families named in the article

DaISy
Dynamic identification records

IFAC
Identification benchmark records

NIST
Statistical reference data

MATLAB
Toolbox and demonstration data

UCI
Machine-learning data repository

DELVE
Learning-method evaluation data

WEKA
Benchmark data collections

These are sources of experiments and regression tasks, not seven physical process models. Availability, licensing and original links must be verified for each selected record.

One selected source record

1. Acquire + identify
Source URL, license, version, checksum and apparatus description

2. Import + inspect
Time base, input/output order, units, missing values and operating range

3. Freeze the split
Separate fitting, selection and final evaluation records before estimation

4. Compare models
Same samples, excitation, horizon, metrics and uncertainty conventions

Dynamic identification

Keep temporal order and independent runs. Prevent overlap of lagged regressors across split boundaries; state the sample period and prediction horizon.

Static regression

State the target, feature units and sampling unit. Preserve grouped dependence. Use a declared validation split and a final held-out test.

Shared safeguards

Fit normalization only on training data. Record exclusions, provenance and random seeds. Report baseline errors with the same evaluation contract.

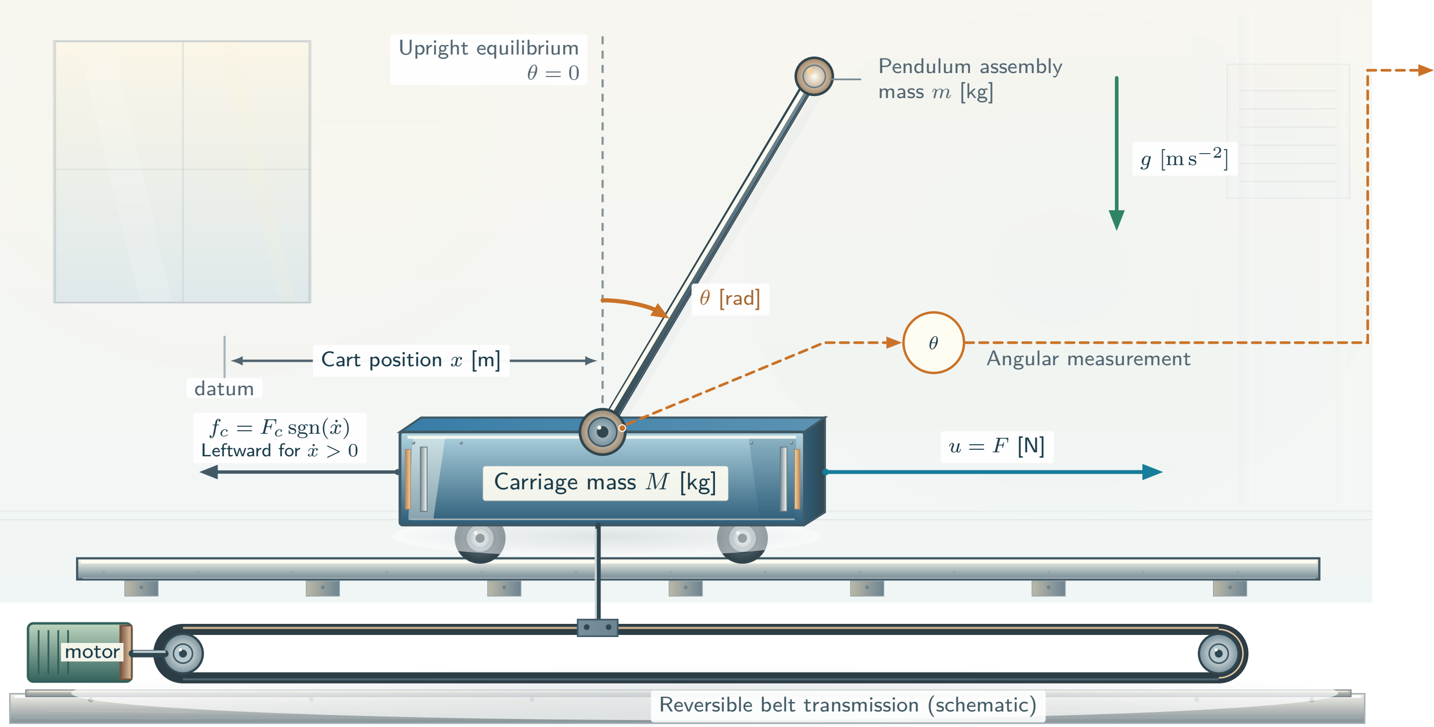
This resource map explains Section 3.2 and the repository import workflow. It does not claim that every historical third-party data set is bundled or still downloadable. Dataset-specific provenance remains necessary before numerical comparison.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Cart and inverted pendulum

Paper section 3.3: *Cart with inverted pendulum*

Belt-driven carriage and freely rotating pendulum


Upright equilibrium
$\theta = 0$
Pendulum assembly
mass m [kg]
g $[\mathrm{m\,s^{-2}}]$
θ [rad]
θ
Angular measurement
Cart position x [m]
datum
$f_c = F_c\,\mathrm{sgn}(\dot{x})$
Leftward for $\dot{x} > 0$
Carriage mass M [kg]
$u = F$ [N]
motor
Reversible belt transmission (schematic)

Control problem

Swing up from the hanging position, balance near the unstable upright state, then position the cart.

Published measurement
Angular position $y = \theta$.

Simulation state order
$[\theta,\ \dot{\theta},\ x,\ \dot{x}]^{\mathsf{T}}$
Units: rad, rad/s, m, m/s.

Nonlinear couplings
Gravity $\sin\theta$, inertial terms $\dot{\theta}^2$, trigonometric coupling and Coulomb friction.

Angular joint loss: $d_{Mf}\dot{\theta}$ [N m].

Input and actuator limit
$u = F$ [N], $|F| \leq F_{\max}$.
The source uses a motor and belt; Eq. (32) takes the resulting force as its input.

Length convention
l [m] is the characteristic length used in Eq. (32). The drawing does not infer a separate rod length, inertia or centre of mass from l.

Reading the drawing
Solid arrows: forces / motion.
Dashed arrows: measurement signals.
Geometry is not to scale.

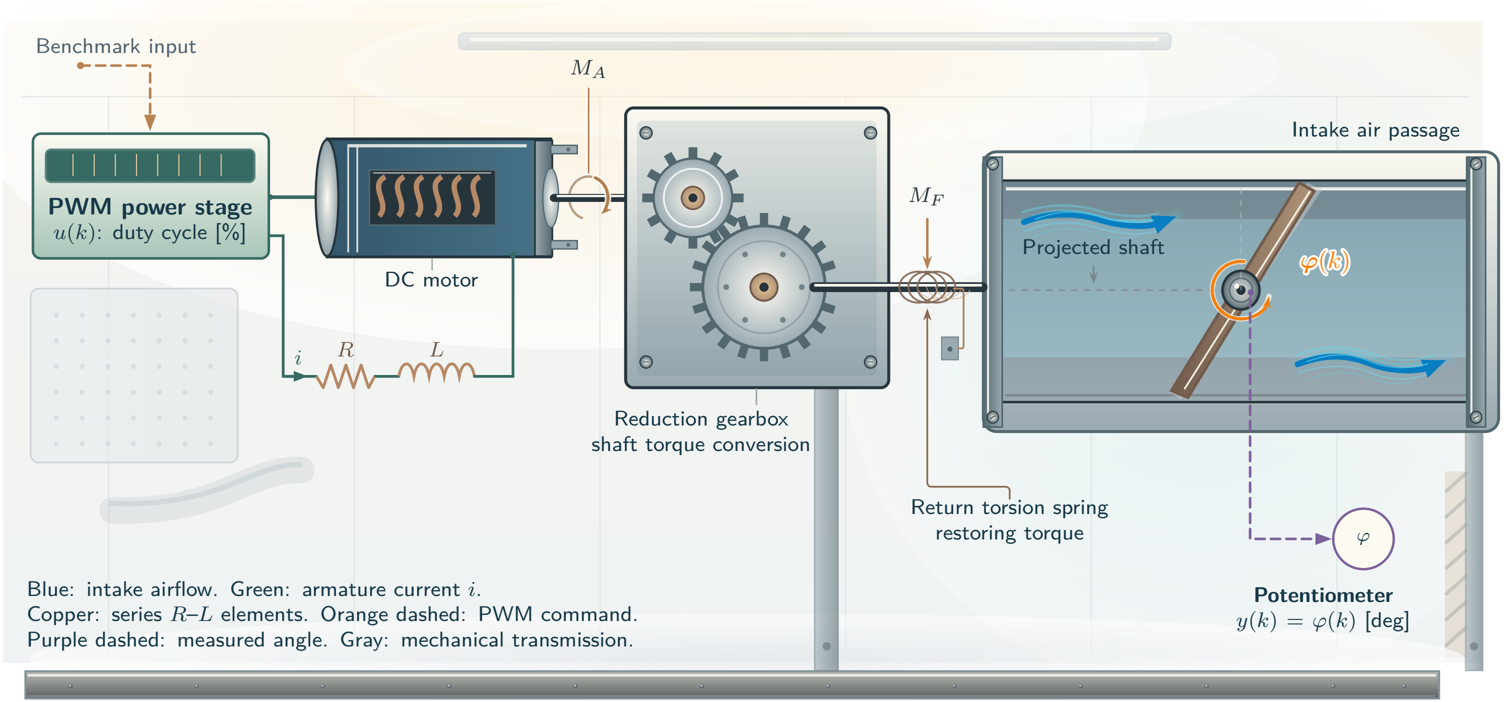
Model scope: Eq. (32) with explicit force saturation and slip friction. No rail-impact or static-sticking model is implied. The educational learning study may declare different friction settings in its case sheet.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Automotive throttle: actuation and identification

Paper section 3.4: *Electro-mechanical throttle*

Physical chain: PWM drive, DC motor, gear box, spring and throttle plate


Benchmark input
M_A
PWM power stage
$u(k)$: duty cycle [%]
DC motor
i
R
L
Reduction gearbox
shaft torque conversion
M_F
Intake air passage
Projected shaft
$\varphi(k)$
Return torsion spring
restoring torque
φ
Potentiometer
$y(k) = \varphi(k)$ [deg]

Blue: intake airflow. Green: armature current i.
Copper: series R–L elements. Orange dashed: PWM command.
Purple dashed: measured angle. Gray: mechanical transmission.

Automotive engine-test-cell rig (illustrative equipment and lighting; not to scale)

Acquisition
Sample interval: $T_0 = 10$ ms.

Input: PWM duty [%].
Output: angle [deg].

Published plate range:
10° closed to 90° open.

Butterfly plate
variable opening

Identification challenge
Friction, nonlinear spring and mechanical compliance can change the input–output response.

No calibrated internal parameters are inferred here.

Training experiment
Multisine excitation; $N = 10\,000$.
Fit dynamic models using the training record only. Keep measured duty and angle in their stated units.

Independent validation
Quasi-APRBS excitation; $N = 2\,500$.
Evaluate one-step and free-run prediction separately; report errors, residuals and model complexity.

Repository evidence
PWA/TS-inspired educational routines are available. Original measured time series have not been recovered; fixtures are synthetic.

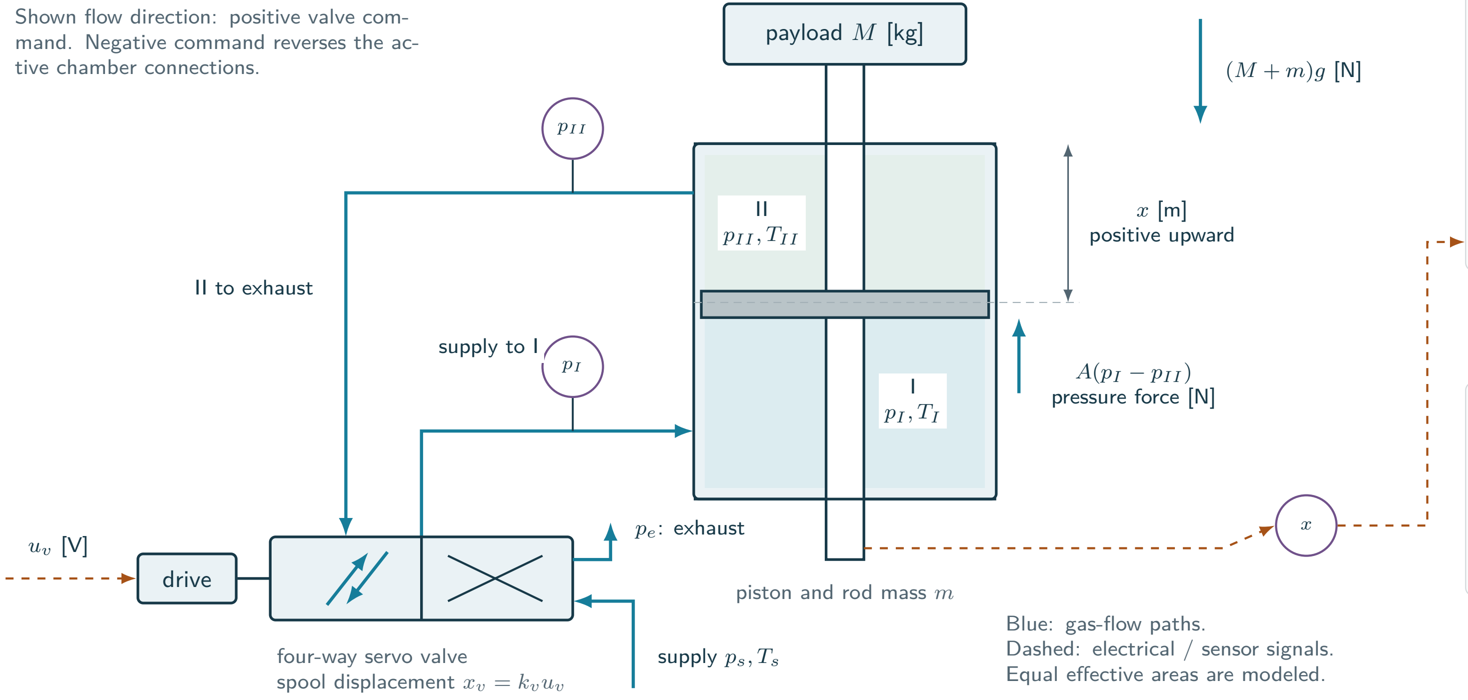
Technology illustration, not a calibrated throttle model or a production assembly drawing. Gear ratio, motor constants, spring law and friction are not inferred. See `THROTTLE_RECOVERY.md` for data access status.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Vertical servo-pneumatic actuator

Paper section 3.5: *Servo-pneumatic actuator*

Two variable-volume chambers, valve flow and a vertical payload

Shown flow direction: positive valve command. Negative command reverses the active chamber connections.



payload M [kg]
$(M+m)g$ [N]
p_{II}
II
p_{II}, T_{II}
x [m]
positive upward
II to exhaust
supply to I
p_I
I
p_I, T_I
$A(p_I - p_{II})$
pressure force [N]
u_v [V]
drive
p_e: exhaust
x
piston and rod mass m
four-way servo valve
spool displacement $x_v = k_v u_v$
supply p_s, T_s

Blue: gas-flow paths.
Dashed: electrical / sensor signals.
Equal effective areas are modeled.

Absolute physical variables

p_I, p_{II}: pressure [Pa].
T_I, T_{II}: temperature [K].
m_I, m_{II}: gas mass [kg].
A: effective piston area [m^2].

$V_I = V_0 + Ax$
$V_{II} = V_0 - Ax$
$p_j = m_j R T_j / V_j$.

Physical variant state

$[x, v, m_I, m_{II}, T_I, T_{II}]^{\mathsf{T}}$

Mass and energy balances evolve each chamber. Flow can choke or reverse. Valve and thermal parameters require calibration.

Vertical force balance

$$(M+m)\dot{v} = A(p_I - p_{II}) \\ -f(v) - (M+m)g - F_{ext}.$$

The physical variant includes gravity explicitly.

Three model meanings

Eqs. (33)–(34): literal survey form.
Local identified models: deviation states with operating offsets removed.
Six-state model: separate physical variant.

Units and payload

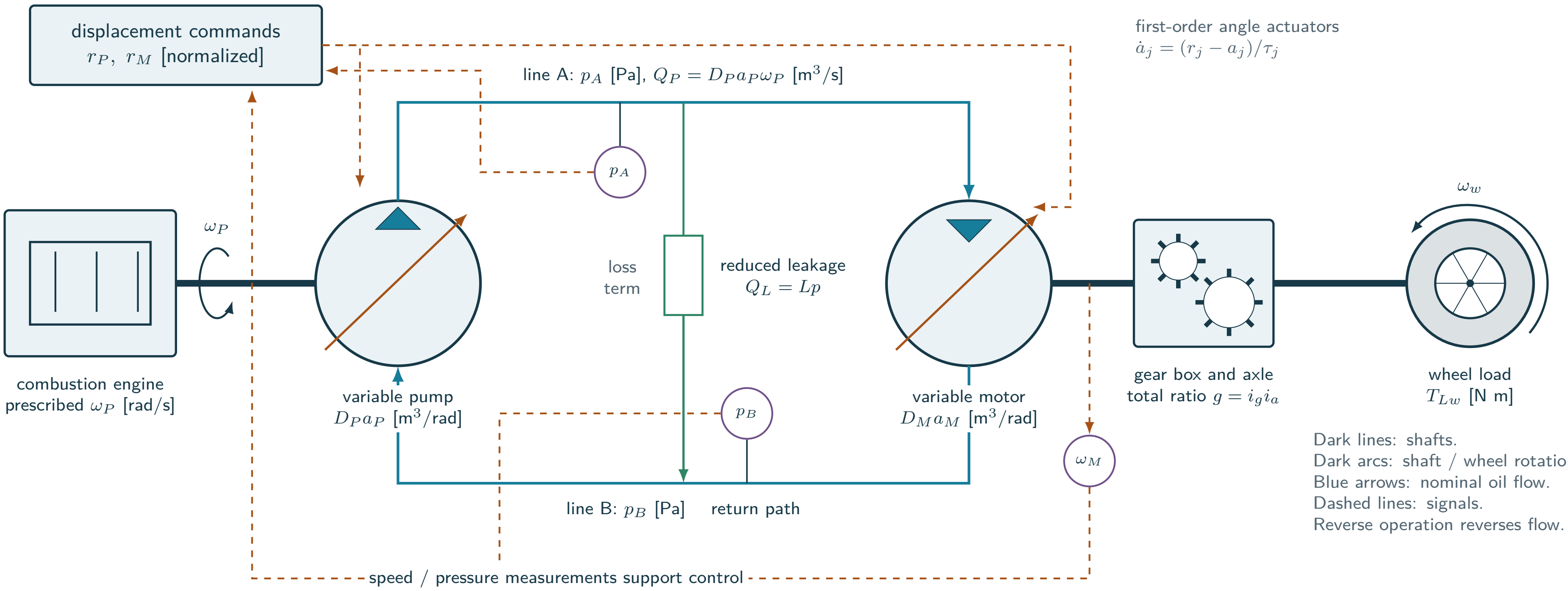
Mass-specific $R = R_u / M_{air}$ [J/(kg K)].
Each physical simulation has constant payload; changing payload in motion needs additional assumptions.

Functional explanatory diagram, not a standardized pneumatic schematic. The survey's gravity omission and gas-constant units are discussed in PNEUMATIC_AUDIT.md. The physical valve closure is an explicitly parameterized teaching variant, not an identified replica of the apparatus.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Hydrostatic drive: closed oil circuit and vehicle load

Paper section 3.6: *Hydrostatic transmission in mobile working machines*

Variable-displacement pump and motor couple hydraulic pressure to shaft speed


displacement commands
r_P, r_M [normalized]
first-order angle actuators
$\dot{a}_j = (r_j - a_j)/\tau_j$
line A: p_A [Pa], $Q_P = D_P a_P \omega_P$ [m^3/s]
p_A
ω_P
ω_w
loss
term
reduced leakage
$Q_L = Lp$
combustion engine
prescribed ω_P [rad/s]
variable pump
$D_P a_P$ [m^3/rad]
p_B
variable motor
$D_M a_M$ [m^3/rad]
gear box and axle
total ratio $g = i_g i_a$
wheel load
T_{Lw} [N m]
ω_M
line B: p_B [Pa]
return path
speed / pressure measurements support control

Dark lines: shafts.
Dark arcs: shaft / wheel rotation.
Blue arrows: nominal oil flow.
Dashed lines: signals.
Reverse operation reverses flow.

SI variant state and storage
$[a_P, a_M, p, \omega_M]^{\mathsf{T}}$, $p = p_A - p_B$.
$C_\Delta \dot{p} = D_P a_P \omega_P - D_M a_M \omega_M - Lp$.
C_Δ [m^3/Pa]: effective capacitance.

Mechanical balance
$J\dot{\omega}_M = \eta D_M a_M p - b\omega_M - T_{Lw}/g$.
Motor torque uses motor displacement.
Displacement per revolution is converted through $D = V/(2\pi)$.

Declared interpretation
Eq. (35) remains a separate transcription. The SI variant requires explicit displacement and capacitance conventions. No electrical actuator calibration is inferred.

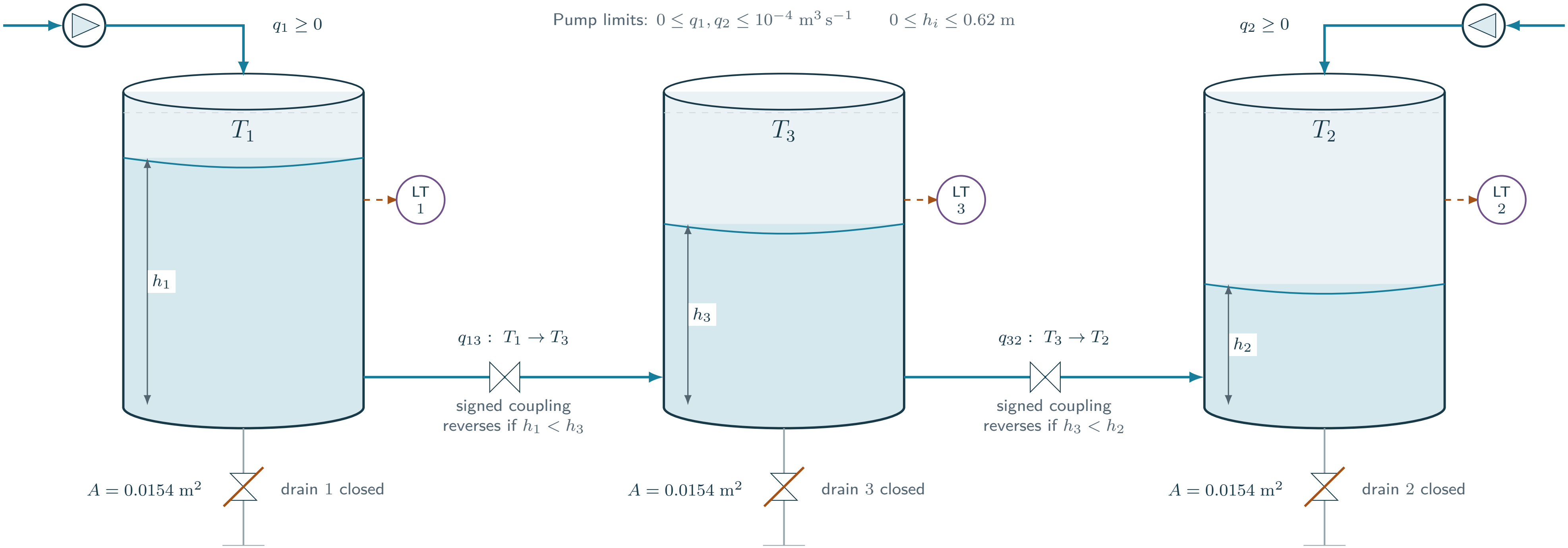
Functional circuit, not a standardized hydraulic drawing. Constant efficiencies are valid for the declared motoring study; no passive four-quadrant, cavitation or relief-valve model is implied. See HYDROSTATIC_SOURCE_REVIEW.md for source conflicts and unresolved calibration.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Three interconnected tanks

Paper section 3.7: *Three-tank-system*

Physical arrangement: T_1–T_3–T_2 | State ordering in the implementation: $[h_1, h_2, h_3]^\mathsf{T}$



$q_1 \geq 0$
Pump limits: $0 \leq q_1, q_2 \leq 10^{-4}\ \mathrm{m}^3\,\mathrm{s}^{-1}$
$0 \leq h_i \leq 0.62\ \mathrm{m}$
$q_2 \geq 0$
T_1
T_3
T_2
LT 1
LT 3
LT 2
h_1
h_3
h_2
$q_{13}:\ T_1 \to T_3$
$q_{32}:\ T_3 \to T_2$
signed coupling reverses if $h_1 < h_3$
signed coupling reverses if $h_3 < h_2$
$A = 0.0154\ \mathrm{m}^2$
drain 1 closed
$A = 0.0154\ \mathrm{m}^2$
drain 3 closed
$A = 0.0154\ \mathrm{m}^2$
drain 2 closed

Blue: liquid flows. Dashed orange: level observations. Crossed valves: zero nominal discharge; drain/fault experiments are explicit extensions.

Hydraulic coupling

$$q_{ij} = a\,\mu\,\operatorname{sgn}(h_i - h_j) \cdot \sqrt{2g|h_i - h_j|}$$

$a = 0.5 \times 10^{-4}\ \mathrm{m}^2$, $\mu = 0.48$.

Nominal level balances

$$A\dot{h}_1 = q_1 - q_{13}$$
$$A\dot{h}_3 = q_{13} - q_{32}$$
$$A\dot{h}_2 = q_2 + q_{32}$$

Time in seconds; levels in metres.

Conservation and operating limit

$$\frac{d}{dt}A(h_1 + h_2 + h_3) = q_1 + q_2.$$

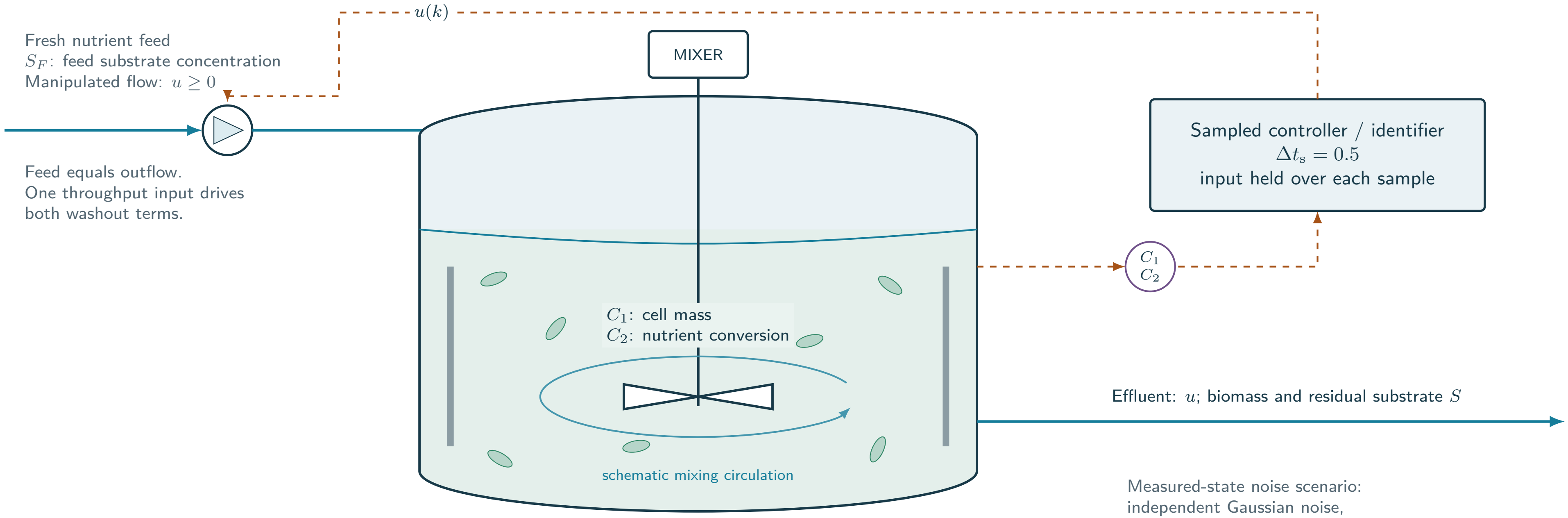
With closed drains, positive pumping rules out a steady equilibrium. Corrected area unit: m^2.

Scope: three-state nominal model, Eq. (36). Vessel heights in this drawing are illustrative. A tracking experiment with continuous pumping requires an explicitly declared outlet extension.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Continuous biochemical reactor

Paper section 3.8: *Continuously operated bio-chemical reactor*

Two nonlinear states | Constant-volume, well-mixed process | Dimensionless model and time



Fresh nutrient feed
S_F: feed substrate concentration
Manipulated flow: $u \geq 0$

Feed equals outflow.
One throughput input drives
both washout terms.

C_1: cell mass
C_2: nutrient conversion

schematic mixing circulation

Sampled controller / identifier
$\Delta t_s = 0.5$
input held over each sample

Effluent: u; biomass and residual substrate S

Measured-state noise scenario:
independent Gaussian noise,
standard deviation 0.01 per state.

Ideal mixing: outlet has the reactor composition.
Cells are symbolic; no aeration or thermal model is implied.

Normalized state definitions

$C_2 = (S_F - S)/S_F, \quad 0 \leq C_1, C_2 \leq 1.$

$r = C_1(1 - C_2)\exp(C_2/\Gamma).$

Nominal $\Gamma = 0.4$, $\beta = 0.02$.

Growth and washout balances

$$\dot{C}_1 = -uC_1 + r$$

$$\dot{C}_2 = -uC_2 + r\frac{1+\beta}{1+\beta-C_2}.$$

No dimensional flow or time scale is supplied.

What makes this a benchmark

Stable and unstable operating regions; limit cycles and a non-unique input–yield relation.

One sampled transition uses **50 Euler substeps**, each of length 0.01.

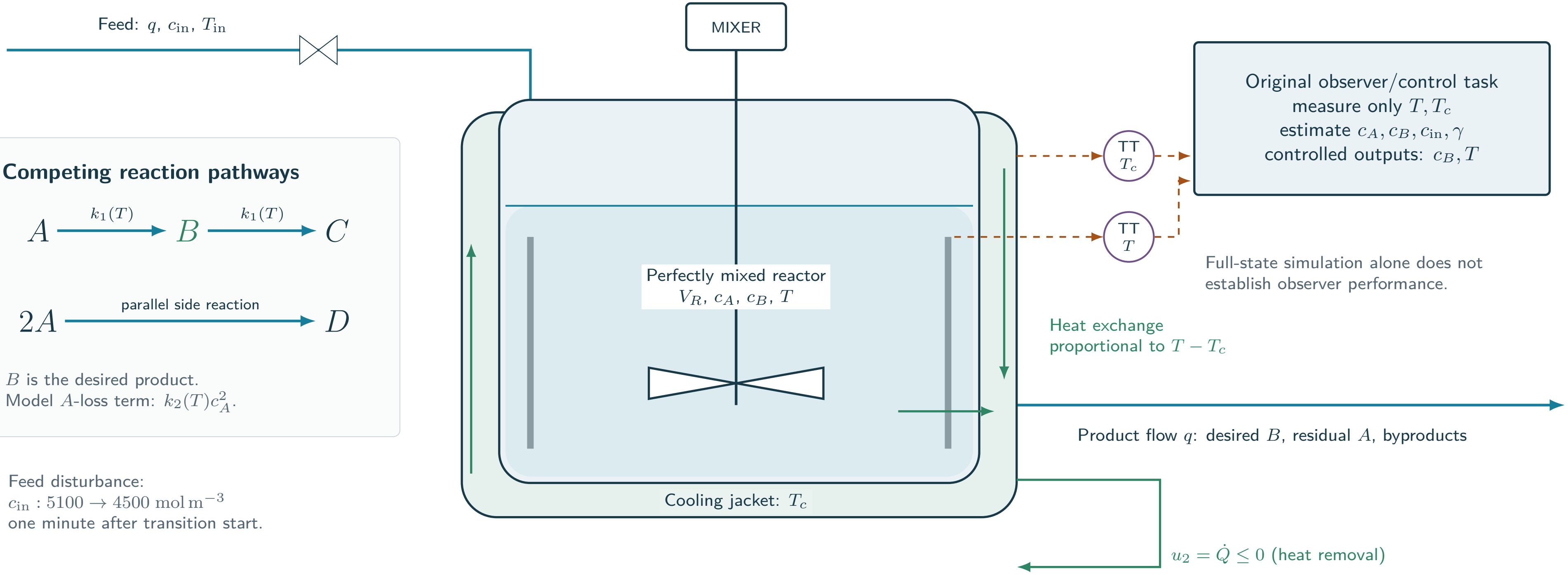
Scope: Eqs. (37)–(38), with dimensionless throughput u. The mixing hardware is explanatory. Equilibria must be recomputed from the equations; the printed biomass at $u = 0.769$ is not consistent with Eq. (37).

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Klatt–Engell jacketed CSTR

Paper section 3.9: *Continuously stirred chemical reactor*

Four states: $[c_A, c_B, T, T_c]^\mathsf{T}$ | Inputs: dilution q/V_R and cooling capacity $\dot{Q}$ | Time: hours



Feed: q, c_{in}, T_{in}

Perfectly mixed reactor
V_R, c_A, c_B, T

Cooling jacket: T_c

Original observer/control task
measure only T, T_c
estimate $c_A, c_B, c_{\text{in}}, \gamma$
controlled outputs: c_B, T

Full-state simulation alone does not establish observer performance.

Heat exchange proportional to $T - T_c$

Product flow q: desired B, residual A, byproducts

$u_2 = \dot{Q} \le 0$ (heat removal)

Competing reaction pathways

$$A \xrightarrow{k_1(T)} B \xrightarrow{k_1(T)} C$$

$$2A \xrightarrow{\text{parallel side reaction}} D$$

B is the desired product.
Model A-loss term: $k_2(T)c_A^2$.

Feed disturbance:
$c_{\text{in}} : 5100 \to 4500\ \text{mol}\,\text{m}^{-3}$
one minute after transition start.

Material balance and kinetics

$$\dot{c}_A = -k_1 c_A - k_2 c_A^2 + (c_{\text{in}} - c_A)u_1$$
$$\dot{c}_B = k_1(c_A - c_B) - c_B u_1$$

$$k_i = k_{i0} \exp[-E_i/(T + 273.15)].$$

Thermal balances

$$\dot{T} = h(c_A, c_B, T) + \alpha(T_c - T) + (T_{\text{in}} - T)u_1$$
$$\dot{T}_c = \beta(T - T_c) + \gamma(1000u_2).$$

h: reaction heat; 1000: MJ to kJ.

Units and input constraints

$5 \le u_1 \le 35\ \text{h}^{-1}$; $u_1 = q/V_R$.
$-8.5 \le u_2 \le 0\ \text{MJ}\,\text{h}^{-1}$.

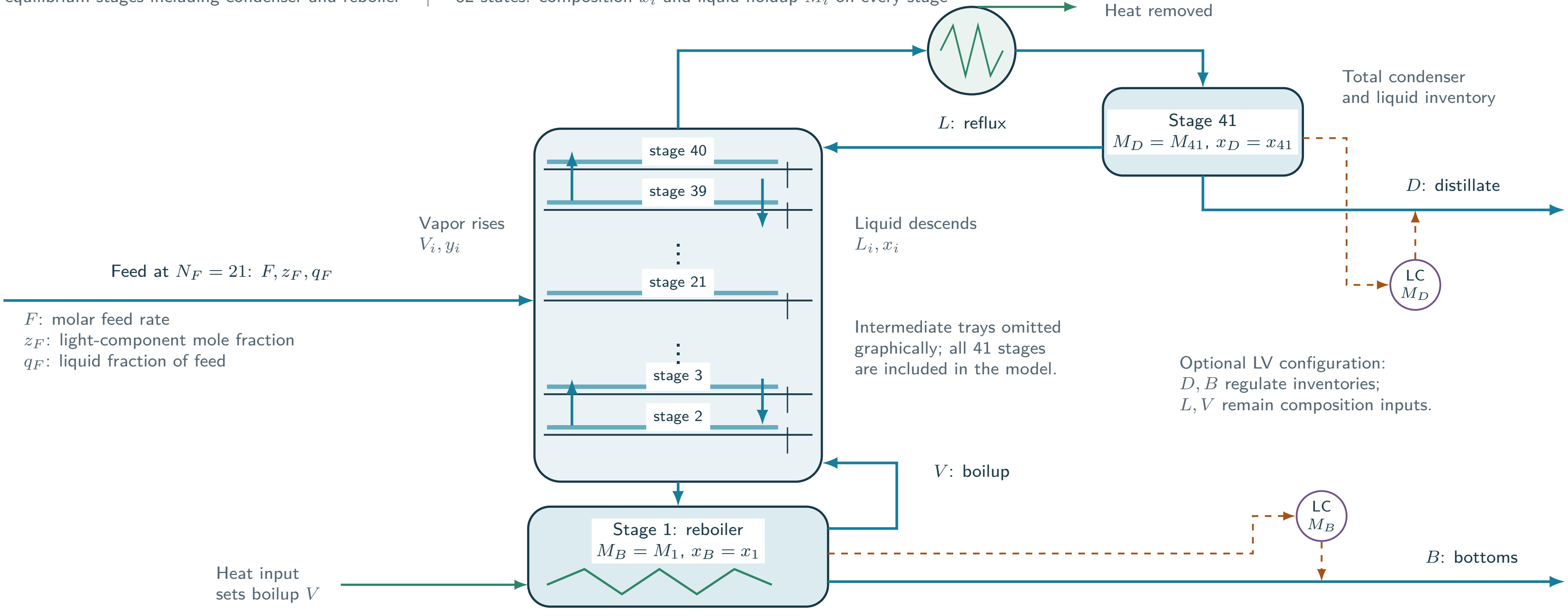
c_A, c_B in $\text{mol}\,\text{m}^{-3}$; T, T_c in $^\circ\text{C}$. Kelvin is used inside Arrhenius rates.

Scope: Eqs. (39)–(43). Celsius resolves the Table 7 temperature-label inconsistency with Eq. (41). The jacket flow arrows illustrate heat-transfer hardware; the model input is cooling capacity, not a specified coolant flow.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Column A: binary distillation

Paper section 3.10: *Continuously operated binary distillation column*

41 equilibrium stages including condenser and reboiler | 82 states: composition x_i and liquid holdup M_i on every stage


Heat removed
Total condenser
and liquid inventory
L: reflux
Stage 41
$M_D = M_{41}$, $x_D = x_{41}$
D: distillate
stage 40
stage 39
stage 21
stage 3
stage 2
Vapor rises
V_i, y_i
Liquid descends
L_i, x_i
LC
M_D
Feed at $N_F = 21$: F, z_F, q_F
V: boilup
LC
M_B
B: bottoms
Stage 1: reboiler
$M_B = M_1$, $x_B = x_1$
Heat input
sets boilup V

F: molar feed rate
z_F: light-component mole fraction
q_F: liquid fraction of feed

Intermediate trays omitted graphically; all 41 stages are included in the model.

Optional LV configuration:
D, B regulate inventories;
L, V remain composition inputs.

Interior-stage conservation

$$\dot{M}_i = L_{i+1} - L_i + V_{i-1} - V_i$$

$$\frac{d(M_i x_i)}{dt} = L_{i+1}x_{i+1} + V_{i-1}y_{i-1} - L_i x_i - V_i y_i.$$

Feed, condenser and reboiler use separate balances.

Thermodynamics and hydraulics

$y_i = \alpha x_i/[1 + (\alpha - 1)x_i]$, $\alpha = 1.5$.
$L_i = L_{0i} + (M_i - M_{0i})/\tau_l$ on trays.

Constant pressure; negligible vapor holdup; constant molar flows in each section.

Interfaces and units

4×4: inputs L, V, D, B; outputs x_D, x_B, M_D, M_B.
Flows: $\mathrm{kmol/min}$; holdups: kmol.

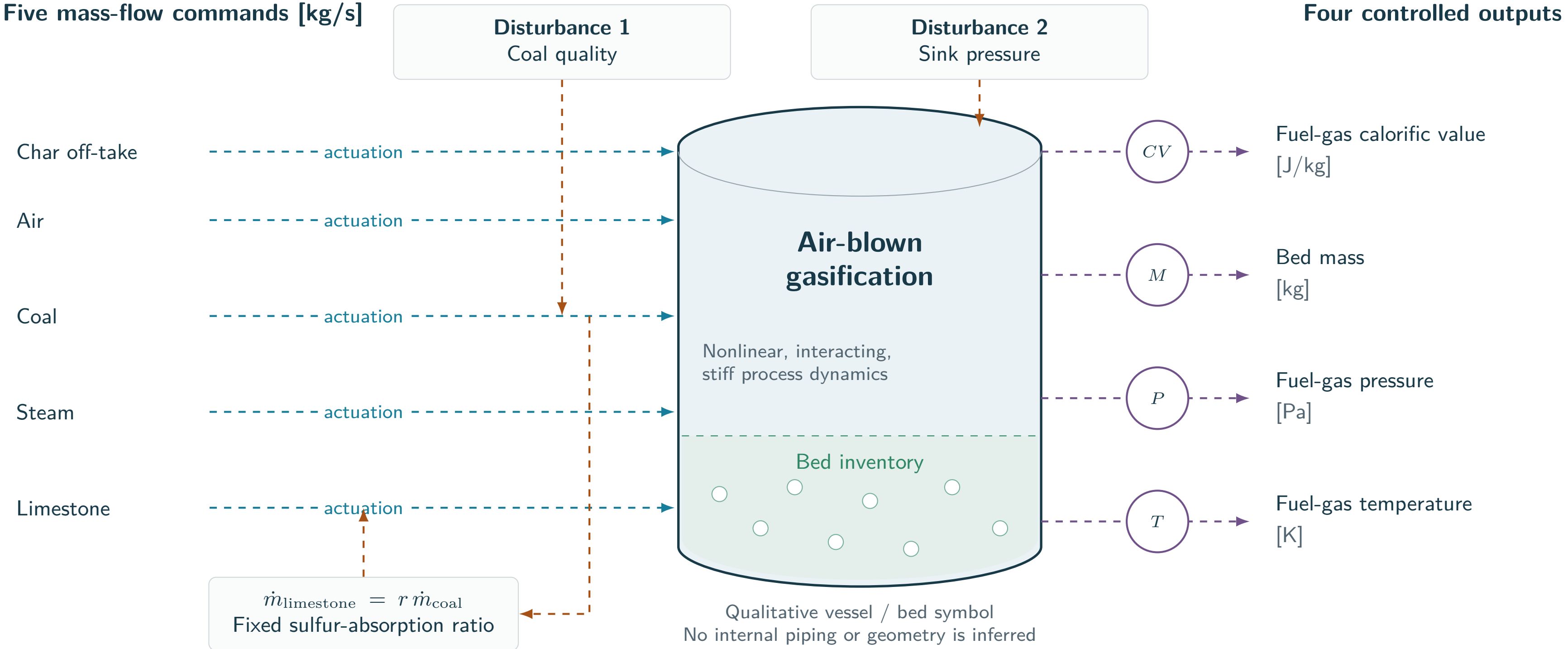
NTNU nominal values: $L_0 = 2.70629$, $V_0 = 3.20629$; $\tau_l = 0.063$ min.

Scope: Column A, Eqs. (47)–(54), aligned with Skogestad's NTNU reference code. Incoming vapor uses V_{i-1}, correcting the printed Eq. (47). Dashed D/B level loops describe the documented LV extension.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

ALSTOM gasifier: nonlinear plant and control interface

Paper section 3.11: *ALSTOM Gasifier Control Benchmark II*


Five mass-flow commands [kg/s]
Disturbance 1
Coal quality
Disturbance 2
Sink pressure
Four controlled outputs
Char off-take
actuation
Air
actuation
Coal
actuation
Steam
actuation
Limestone
actuation
Air-blown gasification
Nonlinear, interacting, stiff process dynamics
Bed inventory
CV
M
P
T
Fuel-gas calorific value [J/kg]
Bed mass [kg]
Fuel-gas pressure [Pa]
Fuel-gas temperature [K]
$\dot{m}_{\text{limestone}} = r\,\dot{m}_{\text{coal}}$
Fixed sulfur-absorption ratio
Qualitative vessel / bed symbol
No internal piping or geometry is inferred

Four independent manipulated flows; the fifth follows coal.
Magnitude and slew limits apply to the actuators.

Official model family

Nonlinear open-loop plant and multivariable PI baseline. The native package contains the dynamics; this drawing specifies the external signal interface.

Challenge experiments

Sink-pressure steps and harmonics; load and coal-quality changes. Compare maximum and integral absolute output errors with the declared bounds.

Interpretation of results

A completed simulation is not a passed control specification. The retained native PI results exceed the temperature band in all six pressure cases.

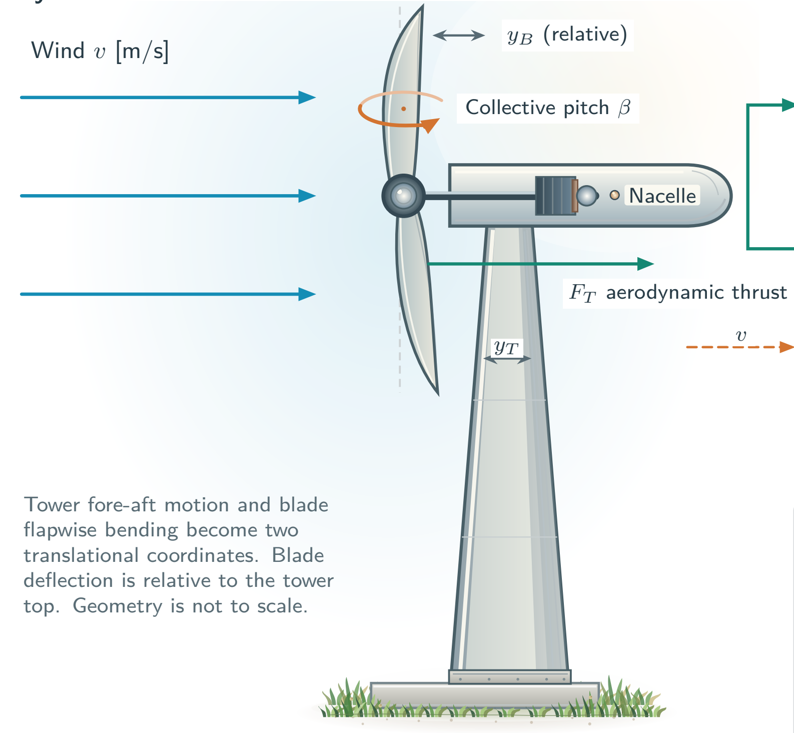
Dashed blue arrows: mass-flow control signals, not physical pipe directions; char is withdrawn from the process. Orange: disturbances and fixed-ratio linkage. Purple: output measurements, not material outlets. Vessel and bed are conceptual; this is not a construction P&ID.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Wind turbine: structure, drivetrain and aerodynamic coupling

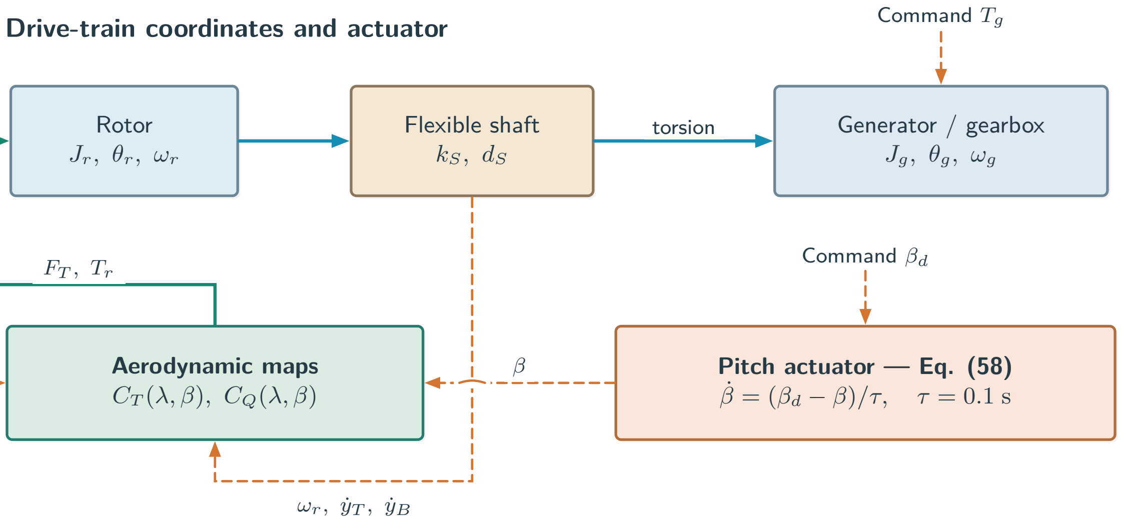
Paper section 3.12: *Wind turbine*

Physical coordinates — schematic side view


Wind v [m/s]
y_B (relative)
Collective pitch β
Nacelle
F_T aerodynamic thrust
y_T

Tower fore-aft motion and blade flapwise bending become two translational coordinates. Blade deflection is relative to the tower top. Geometry is not to scale.

Drive-train coordinates and actuator


Command T_g
Rotor
$J_r,\ \theta_r,\ \omega_r$
Flexible shaft
$k_S,\ d_S$
torsion
Generator / gearbox
$J_g,\ \theta_g,\ \omega_g$
$F_T,\ T_r$
Command β_d
v
Aerodynamic maps
$C_T(\lambda,\beta),\ C_Q(\lambda,\beta)$
β
Pitch actuator — Eq. (58)
$\dot\beta = (\beta_d - \beta)/\tau, \quad \tau = 0.1\ \mathrm{s}$
$\omega_r,\ \dot y_T,\ \dot y_B$

$$v_{\mathrm{eff}} = v - \dot y_T - \dot y_B, \qquad \lambda = \frac{\omega_r R}{v_{\mathrm{eff}}}, \qquad R = 63\ \mathrm{m}$$

$$F_T = \tfrac{1}{2}\rho\pi R^2 C_T v_{\mathrm{eff}}^2, \qquad T_r = \tfrac{1}{2}\rho\pi R^3 C_Q v_{\mathrm{eff}}^2$$

$$M\ddot q + D\dot q + Kq = [F_T, F_T, T_r, -T_g]^{\mathsf{T}}$$

Generator inertia, angle and torque are referred to the low-speed side. The gearbox ratio is absorbed into these quantities; gearbox dynamics are omitted.

Nine-state reduced model

$q = [y_T, y_B, \theta_r, \theta_g]^{\mathsf{T}}$.
$x = [q^{\mathsf{T}}, \dot q^{\mathsf{T}}, \beta]^{\mathsf{T}}$.
Eqs. (55)–(58) retain four mechanical degrees of freedom and first-order pitch.

Control-oriented repository variant

The bundled 2022 ROSCO rotor map supplies the aerodynamic coefficients. Local LQR/PDC gains are newly computed; they do not reproduce Table 12.

High-order verification family

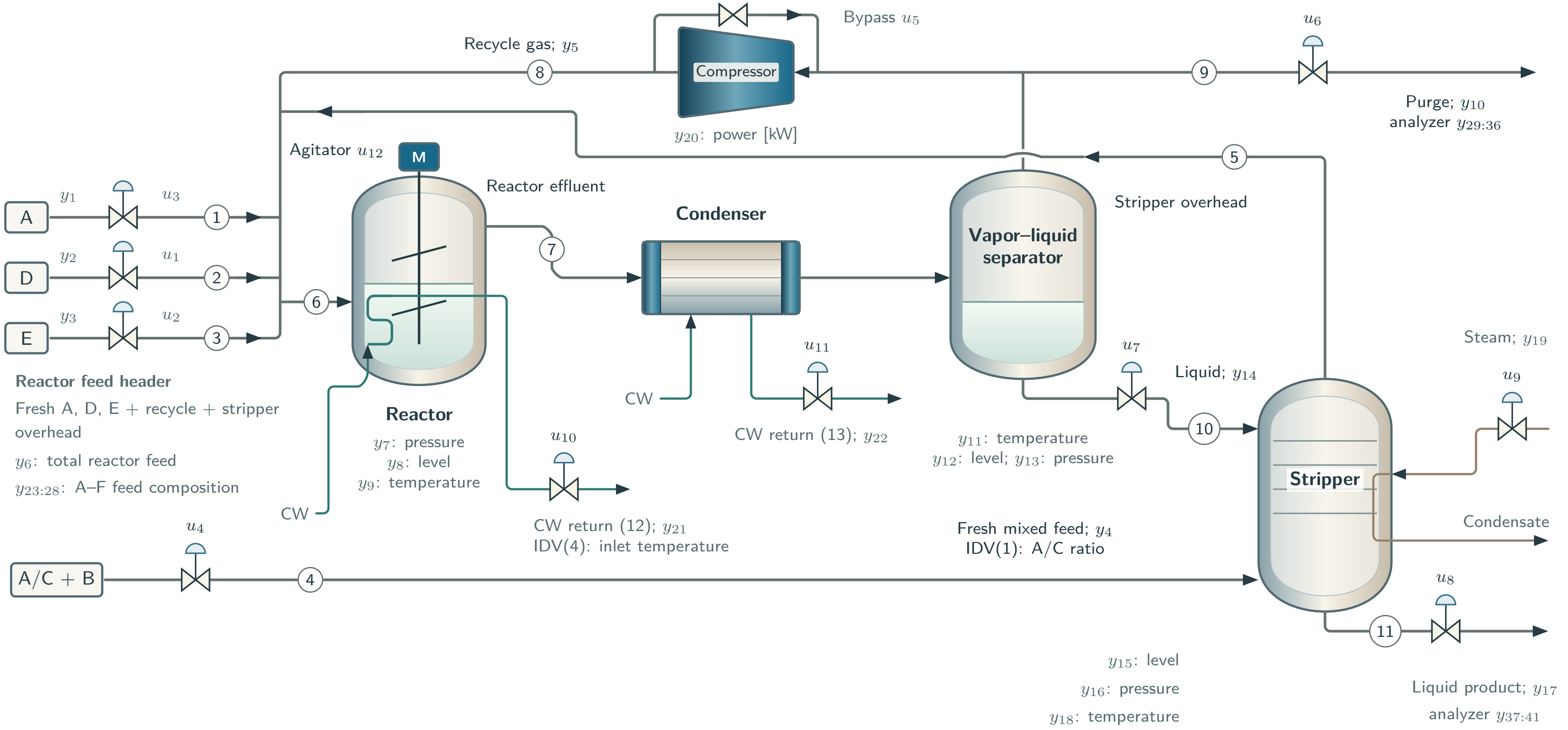
The article also names NREL FAST. The repository executes a later OpenFAST 5.0.0 NREL 5 MW case; it is distinct from this nine-state model.

Blue: mechanical propagation or incident wind. Green: aerodynamic force/torque. Solid orange arc: blade-axis pitch. Orange dashed: model signals and commands. The reduced plant has two commands (T_g, β_d), wind disturbance v, and nine states. FAST/OpenFAST is a separate high-order verification model family.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Tennessee Eastman: process, utilities and instruments

Paper section 3.13: *Tennessee Eastman process control problem*


Recycle gas; y_5
Bypass u_5
Compressor
y_{20}: power [kW]
u_6
8
9
Purge; y_{10}
analyzer $y_{29:36}$
5
Agitator u_{12}
M
Reactor effluent
Stripper overhead
A
D
E
y_1
y_2
y_3
u_3
u_1
u_2
1
2
3
6
7
Condenser
Vapor–liquid separator
Reactor
y_7: pressure
y_8: level
y_9: temperature
Reactor feed header
Fresh A, D, E + recycle + stripper overhead
y_6: total reactor feed
$y_{23:28}$: A–F feed composition
CW
CW
u_{11}
CW return (13); y_{22}
u_{10}
CW return (12); y_{21}
IDV(4): inlet temperature
u_7
Liquid; y_{14}
10
y_{11}: temperature
y_{12}: level; y_{13}: pressure
Steam; y_{19}
u_9
Stripper
Condensate
Fresh mixed feed; y_4
IDV(1): A/C ratio
A/C + B
u_4
4
u_8
11
y_{15}: level
y_{16}: pressure
y_{18}: temperature
Liquid product; y_{17}
analyzer $y_{37:41}$

Chemistry and objective

A, C, D, E react; B is inert; G and H are products; F is a byproduct. Reject disturbances while preserving inventories, product rate and quality.

Model boundary

50 states, 12 commands, 41 measurements, six operating modes. The archived native C revision also defines `NX=50`. Downstream G/H separation is outside the benchmark.

Reading the drawing

Gray: process. Teal: CW. Tan: steam/condensate. T-junctions connect. White bridge: crossing. Finishes and fluid tints are illustrative. $u_i = \mathrm{XMV}(i)$; $y_i = \mathrm{XMEAS}(i)$.

Topology checked against Fig. 13 and the Ricker archive; signal indices checked against `temexd_mod.c`. Original explanatory flowsheet, not a construction P&ID. Instrument labels identify measurements, not a prescribed controller.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

Tennessee Eastman: benchmark signal reference

Paper section 3.13: *Tennessee Eastman process control problem*

12 manipulated variables: XMV

Valve and agitator commands are expressed as percentages.

1 D feed, stream 2
2 E feed, stream 3
3 A feed, stream 1
4 A/C mixed feed, stream 4
5 Compressor recycle bypass
6 Purge, stream 9
7 Separator liquid, stream 10
8 Stripper product, stream 11
9 Stripper steam
10 Reactor cooling-water outlet
11 Condenser cooling-water outlet
12 Reactor agitator command [%]

Source terminology sometimes calls the condenser coolant the separator coolant; it is the same thermal path. Commands and realized valve positions are distinct model quantities.

Representative disturbance paths

IDV(1): A/C feed ratio, constant B.
IDV(4): reactor coolant inlet temperature.
IDV(6): loss of A feed.
IDV(13): kinetics drift.
IDV(14–15): coolant valve sticking.

The repository's archived native runs cover nominal operation, IDV(1) and IDV(4); this diagram adds no new execution evidence.

Continuous measurements: XMEAS(1:22)

1	A feed	kscmh
2	D feed	kg/h
3	E feed	kg/h
4	A/C mixed feed	kscmh
5	Recycle flow	kscmh
6	Reactor feed	kscmh
7	Reactor pressure	kPa gauge
8	Reactor level	%
9	Reactor temperature	°C
10	Purge flow	kscmh
11	Separator temperature	°C
12	Separator level	%
13	Separator pressure	kPa gauge
14	Separator underflow	m^3/h
15	Stripper level	%
16	Stripper pressure	kPa gauge
17	Stripper underflow	m^3/h
18	Stripper temperature	°C
19	Stripper steam	kg/h
20	Compressor power	kW
21	Reactor coolant outlet	°C
22	Condenser coolant outlet	°C

kscmh: thousands of standard cubic meters per hour.

Composition analyzers: XMEAS(23:41)

All concentrations are molar percentages.

Reactor feed, stream 6
$23{:}28 \longleftrightarrow A, B, C, D, E, F$

Purge, stream 9
$29{:}36 \longleftrightarrow A, B, C, D, E, F, G, H$

Product, stream 11
$37{:}41 \longleftrightarrow D, E, F, G, H$

Analyzer sample/hold, noise and delay are part of the simulator. A composition signal is not an instantaneous complete state observation.

Verify the implemented state count
The survey describes 50 original states. The archived native C revision also uses `NX=50`, with 50 derivatives. Its introductory comment says 52, but that comment conflicts with the executable declarations. See the state-count audit; extra diagnostic outputs are not additional continuous states.

From the flowsheet to a learning or diagnostic experiment

Record the simulator revision, operating mode, solver, time base, random seed, initial state, command limits and disturbance schedule. Select measured channels before fitting or tuning. Respect the chronological split and analyzer timing.

Markov/HMM sensor modes in the course are explicitly synthetic overlays. They must not be identified with official process disturbances or with experimentally calibrated equipment-failure probabilities.

Signal labels and units checked against the Ricker/Bathelt archive's documented C interface. State count checked against its executable declarations. The gallery links the archive, source hash and state-count audit.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.